

Armazéns de Dados

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Bibliography

- OLAP Solutions
Building Multidimensional Information Systems (2nd Ed.)
Erik Thomsen
John Wiley and Sons, 2002

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On-Line Analytical Processing (OLAP)

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On-Line Analytical Processing

- Defined by **E.F. Codd**

- ➔ White paper "Providing OLAP (On-line Analytical Processing) to User-Analysts: An IT Mandate", 1993
- ➔ OLAP is "the dynamic synthesis, analysis, and consolidation of large volumes of **multidimensional** data."
- ➔ Codd proposed an arrangement of data in arrays to allow fast analysis – **cubes**
- ➔ **OLAP cubes** can produce an answer in around 0.1% of the time for the same query on a OLTP relational database

- **Complements** OLTP and DW systems

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OLAP Analytical Goals

- Before developing OLAP applications it is crucial to identify the business analytical goals – **Key Performance Indicators (KPIs)**
- KPIs are financial and non-financial metrics** used to provide business users with an **indication of the current and historical performance** of different aspects of the business
- KPIs** examples:
 - Profit
 - Loss
 - Gross margin
 - Customer attrition/churn
 - ...

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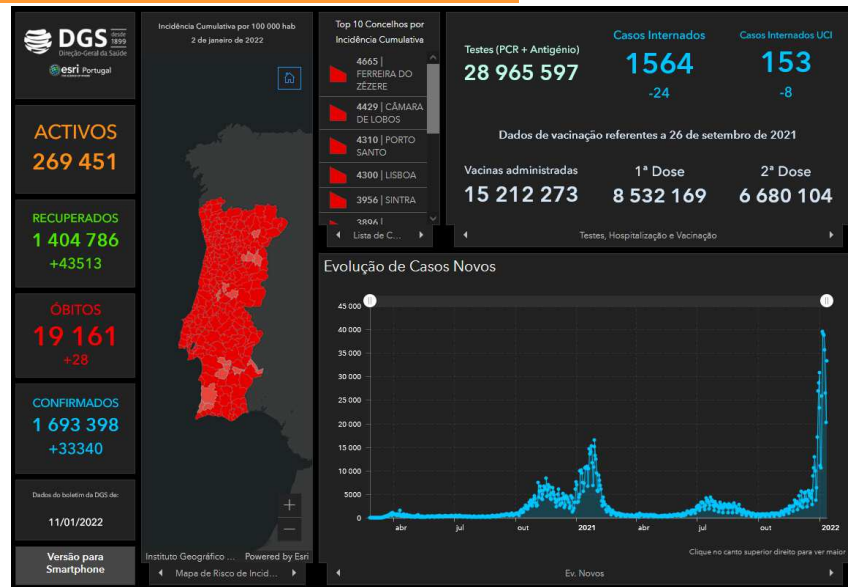
KPI Dashboard Example



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Another KPI Dashboard Example



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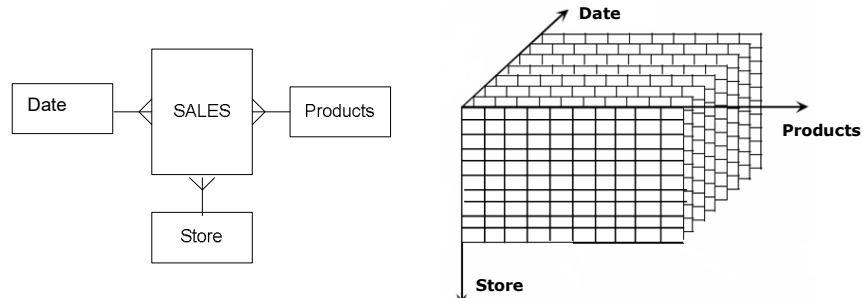
On-Line Analytical Processing

- Approach to quickly provide the answer to **analytical queries** that are multidimensional in nature
 - Usually the data comes from the DW
- Analytical queries** involve inter-row calculations, time series analysis, and access to aggregated historical and atomic/aggregated current data
- Multidimensional data model** allowing analytical ad-hoc queries with fast execution time
- Output of an OLAP query** is typically displayed in a **matrix** (or pivot) format:
 - Dimensions** – rows and columns of the matrix
 - Measures** (facts) – values of each cell

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OLAP Functionality

- In the core of any OLAP system is the concept of an **OLAP cube** also called a **multidimensional cube** or a **hypercube**



- **Hypercube** consists of numeric facts called **measures** which are categorized by **dimensions**
- Cube is typically created from a star schema of tables in a relational database
 - **Measures** are derived from the records in the **fact table**
 - **Dimensions** are derived from the records in the **dimension tables**

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OLAP Applications

- **Multidimensional** data structure
- Automatic **roll-ups** and **drill-downs**
 - No programming required
- Support **complex calculations**
 - Sales forecasting, moving averages, growth percentage, etc.
- **Time intelligence**
 - Comparison of performance between different time periods

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Kinds of On-Line Analytical Processing

▪ MOLAP (Multidimensional OLAP)

- “Classic” form of OLAP, sometimes referred to as just OLAP
- Data is stored in a special, usually proprietary, data structure

▪ ROLAP (Relational OLAP)

- Works directly with relational databases
- Tools access the DW directly

▪ HOLAP (Hybrid OLAP)

- Takes synergies from ROLAP and MOLAP
- Divides data between relational and specialized storage
 - ♦ Uses relational tables to hold larger quantities of detailed data
 - ♦ Uses specialized storage for smaller quantities of more-aggregate or less-detailed data

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Multidimensional OLAP

- Stores data into an **optimized multidimensional array storage**, usually a three or higher dimensional array (**hipercube**)
- **Requires the pre-computation and storage of data in the hipercube** for fast end-user response
 - Operation known as **processing**
 - ♦ Expensive
 - ♦ Makes loading slower
- Examples: Cognos Powerplay; Hyperion ESSbase; SQL Server Analysis Services

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MOLAP Advantages

- Delivers **better performance** due to specialized indexing and storage optimizations
 - Array model provides **natural indexing** – easy and quick to access an array's position
- **Automated computation of higher-level aggregates** of data
- Does not need to have a **permanent connection** to the underlying DW

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MOLAP Disadvantages

- **Data loading** into these structures can be **lengthy**, especially of large data volumes
 - Must be **periodically updated** to remain current
 - ➔ **Solution: incremental processing** – only the data which is new or has been updated, instead of reprocessing the entire hypercube
- Stores a copy of the data at OLAP server and so **requires additional storage**

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Relational OLAP

- **Does not require the pre-computation of data** (as in MOLAP)
 - Creates multidimensional views from a relational database
- ROLAP tools **access the data in a relational database** and generate SQL queries to retrieve data when an end-user requests it
- **Relational database** must be carefully **designed for ROLAP use** – a database which was designed for OLTP will not function well as a ROLAP database
 - Data must be stored in a **dimensional data model**
 - **It is possible to create additional database tables** (*aggregations*) which summarize the data at any desired combination of dimensions
- Examples: SAP BW; MicroStrategy; SQL Server Analysis Services

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ROLAP Advantages

- Uses an **underlying RDBMS**, rather than a specialized multidimensional data structure
 - **Better scalability** since is able to handle very high volumes of data
 - **Maturity** of the RDBMS technology
- **Frequent updates** are not a problem
- It has **less storage requirements**, since it does not make a copy of the data (as in MOLAP)

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ROLAP Disadvantages

- As the data structure is relational, **SQL must be used to access the detailed records**
 - ROLAP engine must perform additional work
 - SQL generation may be costly
- **Slower data retrieval** than with MOLAP tools

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Hybrid OLAP

- **Combination** of MOLAP and ROLAP that mixes the **best** of both
 - Can process quickly and scale well
- Combines the **performance and functionality of the multidimensional database** with the **ability to access directly detailed data on relational databases**
 - Stores the **aggregations** that exist within a **multidimensional** schema, leaving the **most detailed data** in **relational** form
- Most commercial OLAP tools **now support** a hybrid OLAP approach
- Examples: Holo; Oracle Advanced Analytic Services; SQL Server Analysis Services

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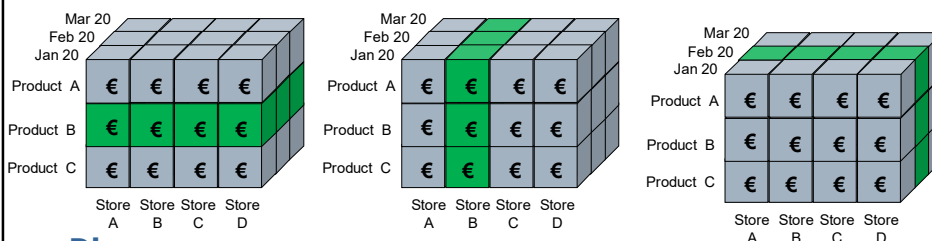
OLAP Basic Operations

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Slice & Dice

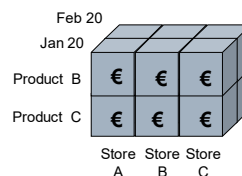
■ Slice

- Subset of multidimensional data
- Defined through the selection of specific dimension attribute values



■ Dice

- Extracts a sub-cube from the original cube
- Can be combined with slicing

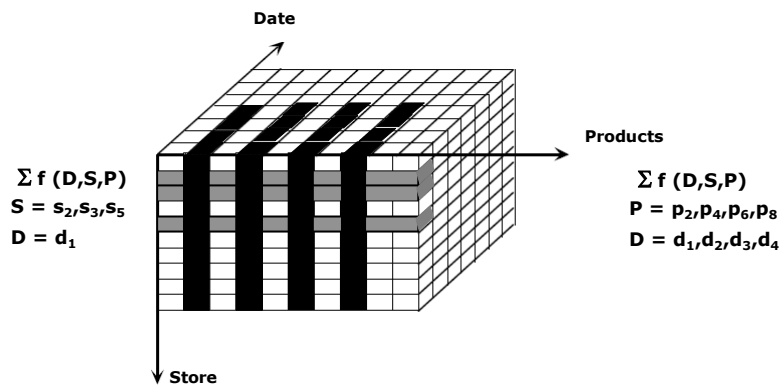


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Aggregations

- Built by **restricting the values of certain dimension attributes** and **aggregating data along them**

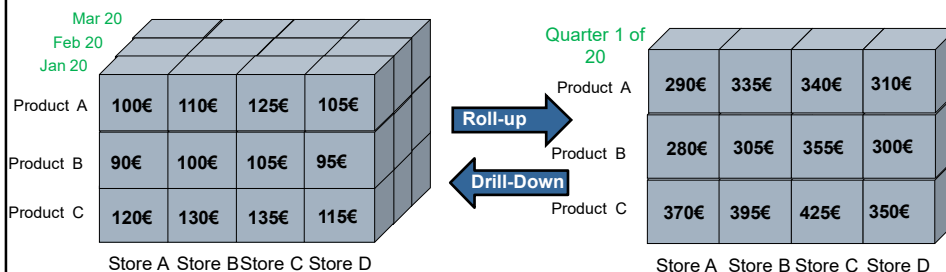


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Drill-Down & Roll-Up

- Drill-down** means to move **from summary information to more detailed data** by focusing on something
- Roll-up** means to move **from more detailed data to summary information** by abstracting of something



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Pivot Tables

Data Table		
Model	Color	Sales
Van	Blue	6
Van	Red	5
Van	White	4
Coupe	Blue	3
Coupe	Red	5
Coupe	White	5
Sedan	Blue	4
Sedan	Red	3
Sedan	White	2

Pivot Tables are especially well-suited for **taking enormous amounts of data and summarizing that data into useful reports**

Pivot Table 3 x 3

	Color		
Model			

Pivot Table				
Model/Color	Blue	Red	White	Total
Van	6	5	4	15
Coupe	3	5	5	13
Sedan	4	3	2	9
Total	13	13	11	37

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One More Dimension ...

Data Table			
Market	Model	Color	Sales
Europe	Van	Blue	6
America	Van	Red	5
Europe	Van	White	4
Europe	Coupe	Blue	3
America	Coupe	Red	5
Asia	Van	Red	3
America	Coupe	White	5
America	Van	Blue	3
Europe	Coupe	Red	5
Asia	Coupe	White	5
Asia	Sedan	Blue	4
America	Sedan	Red	3
Europe	Sedan	White	2

Market

Color		
Market		

Color

Model

Market	Model	Color			Total
		Blue	Red	white	
America	Coupe	0	5	5	10
	Sedan	0	3	0	3
	Van	3	5	0	8
	Total	3	13	5	21
Europe	Coupe	3	5	0	8
	Sedan	0	0	2	2
	Van	6	0	4	10
	Total	9	5	6	20
Asia	Coupe	0	0	5	5
	Sedan	4	0	0	4
	Van	0	3	0	3
	Total	4	3	5	12
Total		16	21	16	53

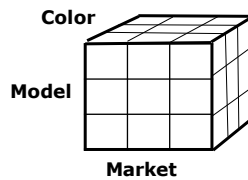
Pivot Table 3 x 3 x 3

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Rotate Data Cube

- Allow the user to **view the same data from a different perspective**



		Market			
Model	Color	America	Europe	Asia	Total
Coupe	Blue	0	3	0	3
	Red	5	5	0	10
	White	5	0	5	10
	Total	10	8	5	23
Sedan	Blue	0	0	4	4
	Red	3	0	0	3
	White	0	2	0	2
	Total	3	2	4	9
Van	Blue	3	6	0	9
	Red	5	0	3	8
	White	0	4	0	4
	Total	8	10	3	21
Total		21	20	12	53

Pivot Table 3 × 3 × 3